A QUASI-FINITE SEPARATED MORPHISM OF ALGEBRAIC SPACES IS QUASI-AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement includes the displayed triangle with the normalization; the proof produces the open immersion statement for f' and then the quasi-affine conclusion in the last sentence.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- A locally quasi-finite separated morphism of algebraic spaces is representable. Lemma lemma-locally-quasi-finite-separated-representable, used at proof lines 11064-11065. Makes f representable so that the factorization lemma applies.
- The factorization of a representable finite type separated morphism through its normalization. Lemma lemma-finite-type-separated, used at proof lines 11065-11069. Produces the open subspace U' with X -> U' an isomorphism.
- Permanence of quasi-compactness for f'. Lemma lemma-quasi-compact-permanence, used at proof lines 11069-11071. Shows that f' is quasi-compact.
- The definition of quasi-affineness for morphisms of algebraic spaces. used at proof lines 11072-11075. Turns the description of the fibre products over affine schemes into the conclusion.

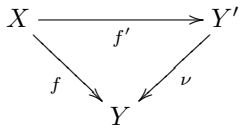
Source reference. Original lemma 0ABR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13029–13029. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0418: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5283–5283. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4688–4688. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is quasi-finite and separated. Let Y' be the normalization of Y in X. Picture:*



Then f' is a quasi-compact open immersion and ν is integral. In particular f is quasi-affine.

Proof. By Lemma 0418 the morphism f is representable. Hence we may apply Lemma 0ABR. Thus there exists an open subspace $U' \subset Y'$ such that $(f')^{-1}(U') = X$ (!) and $X \to U'$ is an isomorphism! In other words, f' is an open immersion. Note that f' is quasi-compact as f is quasi-compact and $\nu : Y' \to Y$ is separated (Lemma 03KS). Hence for every affine scheme Z and morphism $Z \to Y$ the fibre product $Z \times_Y X$ is a quasi-compact open subscheme of the affine scheme $Z \times_Y Y'$. Hence f is quasi-affine by definition. □